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Multi-sensor information fusion based on machine learning for real applications in human activity recognition: State-of-the-art and research challenges

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ABSTRACT

This paper firstly introduces common wearable sensors, smart wearable devices and the key application areas. Since multi-sensor is defined by the presence of more than one model or channel, e.g. visual, audio, environmental and physiological signals. Hence, the fusion methods of multi-modality and multi-location sensors are proposed. Despite it has been contributed several works reviewing the stateoftheart on information fusion or deep learning, all of them only tackled one aspect of the sensor fusion applications, which leads to a lack of comprehensive understanding about it. Therefore, we propose using a more holistic approach in order to provide a more suitable starting point from which to develop a full understanding of the fusion methods of wearable sensors. Specifically, this review attempts to provide a more comprehensive survey of the most important aspects of multi-sensor applications for human activity recognition, including those recently added to the field for unsupervised learning and transfer learning. Finally, the open research issues that need further research and improvement are identified and discussed.

1. Introduction

With the rapid development of artificial intelligence, big data, 5G, the Internet of Things and other technologies, more and more intelligent sensors are changing human lifestyle and even social structure with new products, new technologies, new industries and new models, and wearable devices are one of them. Due to the limitations of computer vision, such as space–time limitations, easy invasion of user privacy, high energy consumption and so on, the emerging sensor based methods have gradually become the focus of attention. The high computational power, small size, and low cost of sensors have led to the development of wearable sensing devices to interact as part of everyday life, making it possible to detect human activities and emotions at any time [1].

Wearable sensors can measure parameters including: pulse, blood oxygen saturation, breathing monitoring, gesture recognition, acceleration and inertial monitoring during human movement, blood pressure, body temperature, ion salts (such as calcium, sodium, potassium and oxides), lactic acid content, cholesterol and urea, etc. With advances in microelectronics and communication technologies, all these parameters can be measured and monitored remotely and transmitted autonomously to nearby service stations (e.g. hospitals, police stations, service centers, etc.) in case of emergency. Wearable sensors are not only flexible and easy to operate, but also durable and less costly.

Considering the portability and comfort of sensor wearing, some studies have used single sensor for action recognition [2,3]. However, single sensor can only acquire local motion information and have low recognition accuracy for complex movements. In recent years, researchers have tried various fusion strategies in order to make full use of data, features and classifiers for effective health and activity monitoring. But we need to choose the number of sensors carefully, because in real life applications, using a small set of sensors can make the process of identifying the system faster and convenient. In the case of wearable sensors, carrying multiple sensors is difficult for users. The number of sensors plays an important role because of personal space and comfort levels [4].

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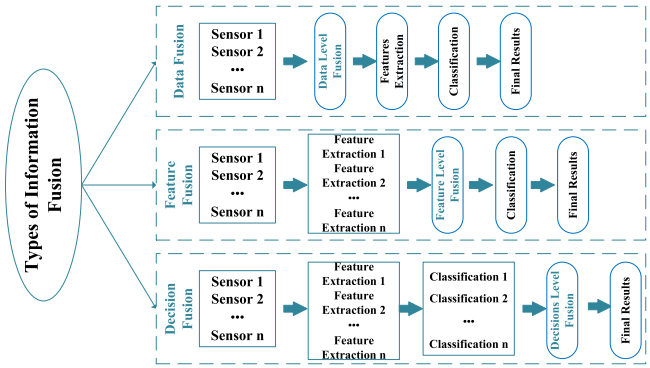

Fig. 1. Classification of Information Fusion.

In terms of the abstraction level of data processing, multi-sensor fusion is usually divided into three categories: data-level fusion, feature-level fusion, and decision-level fusion [5]. The representations of information fusion are depicted in Fig. 1. The key technologies studied in data-level fusion are mainly the design and implementation of denoising, feature extraction, data classification, and data compression [6]. Data fusion involves the integration of data collected by multiple mobile devices and wearable sensor devices to improve the reliability, robustness and generalization of the recognition system. When the system involves multiple homogeneous sensors measuring the same physical phenomenon, sensor data can be fused directly; in contrast, data generated from heterogeneous sources cannot be fused directly and feature-level or decision-level fusion techniques must be employed.

Feature-level fusion refers to the fact that feature sets extracted from multiple data sources (generated by different sensor nodes or by a single node equipped with multiple physical sensors) can be fused to create a new high-dimensional feature vector that represents the input to the classification/pattern recognition step [7]. Feature fusion is commonly used to characterize data measurement separation signals. In this context, features extracted from heterogeneous or homogeneous sensors at different locations are combined using machine learning algorithms such as support vector machines (SVM), decision trees (DT), and hidden Markov models (HMM) to distinguish data into higher levels of abstraction. In addition, automatic feature representation using deep learning to solve spatial and temporal dependence problems has become a frontier research area for sensor-based human activity recognition (HAR) [8].

Decision-level fusion is the process of selecting or generating a hypothesis from a set of hypotheses generated from the individual (local, usually weak) decisions of multiple sensors [9]. The decision-level fusion output is a unique decision obtained from the local decisions of several homogeneous or heterogeneous sensors, and the high-level decision is realized by using the information extracted to a certain level through the preliminary sensor data or feature level processing. Guo et al. [10] proposed a new type of wearable sensor system that recognizes human daily activities by fusing multiple sensor nodes. The process of this method is as follows: firstly, features are extracted from each sensor node, and then dimension reduction is carried out through generalized discriminant analysis (GDA) to ensure the real-time performance of activity recognition; secondly, the reduced features are classified with the multi-class relevance vector machines (RVM); finally, considering that different sensor nodes can provide heterogeneous and complementary information about human activities, individual classification results are fused at the decision level. Chen et al. [11] proposed a human gesture recognition framework based on the information fusion of a three-axis accelerometer and a multi-channel electromyography (EMG) sensor, which automatically detects the start and end points of meaningful gesture segments through the strength of the EMG signal, and uses decision tree and hidden Markov model methods are used as a decision-level fusion to obtain the final classification.

In the process of information fusion, the features extracted from heterogeneous sensors or homogeneous sensors at different locations or use methods based on probability and statistics to reduce errors; or use machine learning algorithms to combine to differentiate the data to a higher level Hierarchy; or use the automatic feature representation based on deep learning to solve space and time dependent problems. Liu et al. [12] proposed a driving behavior monitoring and analysis method based on motion capture and artificial intelligence technology, which can identify different driving behaviors as normal or abnormal driving behaviors through multiple miniature inertial measurement units. A gradient descent method is used to fuse the original data and continuously update the driver's posture information. Also, in order to accurately identify the bilateral arm movements under different paths, the traditional machine learning-based approach is compared with a deep neural network approach based on joint angle sequences. In the experiment, the recognition rate of the two methods is above 99%, and the algorithms performed satisfactorily in real scenarios. Huynh et al. [13] combined machine learning algorithm with deep learning algorithm to construct an intermediate fusion framework for human activity recognition. It divides all kinds of sensor data into two parts, one is manual feature extraction, the other is coding the data into image information for deep learning model, and then normalizes the feature, and fuses manual feature and depth feature through multi class support vector machine classifier. The results show that this fusion strategy can significantly improve the recognition accuracy of the system, and is superior to the human activity recognition algorithm using machine learning or deep learning alone.

In recent years, data fusion and multi-classifier systems have been widely studied and applied in human activity recognition. Multi-sensor data fusion technology has a mature foundation and has provided satisfactory performance in many activity subjects. Some surveys have also made a good summary of it from the perspective of the technological level of sensing, feature and learning fusion. However, little work has been systematically investigated as to whether wearable device can be better applied in the real life environment. For further analysis, it is necessary to review and summarize these studies for in-depth analysis. Table 1 presents a summary of recent reviews on information fusion and deep learning. Antar et al. [4] made a list of sensing devices, sensors, and applications that can be used to collect activity data, and provided necessary information about some benchmark datasets. Gravina et al. [14] discussed clear motivations and advantages of multi-sensor data fusion and particularly focused on physical activity recognition. Nweke et al. [15] focused on the feature extraction of human activity recognition and the description of multi-classification system. The reviews failed to cover substantial new researches proposed in recent years in deep learning fusion. Li et al. [16] discussed the application fields and recognition methods of multi-user identification. Zhang et al. [17] surveyed the emotion recognition methods based on multi-channel EEG signals as well as multi-modal physiological signals. Singh et al. [18] provided a comprehensive technical insight into the existing fall detection system, and classified various approaches and the challenges encountered during implementation. Fortino et al. [19] defined a multi-sensor data fusion architecture that is effective in supporting collaborative real-time data processing and analysis among collaborative body sensor networks (CBSNs) for classification and detection of joint events.

We have mainly reviewed the important research papers published in the field in the last five years, with nearly half coming from 2019, 2020 and 2021. The main focus was papers from the most reputed publishers such as Elsevier, IEEE, MDPI, ACM, Springer, and Nature. Based on the keywords, we collected and analyzed more than a thousand related literatures. Among these literatures, over 300 most influential articles were selected and discussed in depth in three aspects: sensor fusion, algorithms and future challenges. Among them. there are 49

papers from the year 2021, 63 papers from the year 2020, 27 papers from the year 2019, 32 papers from the year 2018, and 47 papers from the year 2017. The most keywords used for search criteria for this review paper are ("wearable sensors"), ("human activity recognition"), ("information fusion"), ("information fusion" and "decision-level fusion"), ("information fusion" and "future-level fusion"), ("information fusion" and "data-level fusion"), ("deep learning"), ("machine learning"), ("kalman filter"), ("support vector machine"), ("k nearest neighbor"), ("neural network"), ("deep learning" and "architectures"), ("deep learning" and "convolutional neural network" or "recurrent neural network" or "long short-term memory" or "gated recurrent unit network"), ("deep learning" and "CPU" or "GPU"), ("deep learning" and "transfer learning"), ("deep learning" and "imbalanced data"), ("deep learning" and "interpretability of data").

Our review includes more comprehensive sensor studies to cover human activities. Specifically, we offered a comprehensive review of fusion of modal-location and multi-modal sensors, successful application of wearable device information fusion in various fields, feature fusion with handcrafted features and deep learning fusion for automatic feature representation. Furthermore, we summarized future challenges and solutions, including unsupervised learning, transfer learning, data augmentation, and attention mechanisms. Compared with existing surveys, the main contributions of this work can be summarized as follows:

- Different types of sensors and their applications are introduced.
- The information fusion methods of wearable sensors are described in detail from the perspectives of multi-location and multi-modal.
- Summarized the traditional information fusion methods and expanded the role of machine learning and deep learning in information fusion
- Proposed new deep learning methods, such as transfer learning and data augmentation.
- Pointed out the current challenges (limitations) and future researches of deep learning in human activity recognition.
- In addition, some key issues are analyzed and compared in the form of tables and pictures, including the emphasis of previous reviews, accuracy of different fusion methods, advantages and disadvantages, application of sensors and challenges.

The remainder of this paper is organized as follows: Section 2 discusses common wearable sensors, wearable smart mobile devices and main application areas of information fusion. Section 3 explores the multi-location and multi-modal sensor fusion methods for human activity recognition. Section 4 reviews information fusion algorithms based on mobile and wearable sensors, including statistical and probabilistic-based approaches, machine learning algorithms, and deep learning feature representations. Section 5 provides brief research challenges that require further improvement and Section 6 concludes the view.

2. Common wearable devices and main application areas

2.1. Common wearable sensors

2.1.1. Inertial sensors

At present, inertial sensors [31] such as accelerometers, gyroscopes, and magnetometers are widely used for human activity recognition. For example, Majumder et al. [32] proposed a three-stage sensor fusion algorithm using inertial sensors for human health monitoring and assessment in the home. The gradient descent method is used to estimate the gyroscopic drift and subtract it from the accumulated gyroscopic data to obtain real-time orientation information. Weygers et al. [33] proposed a novel joint kinematics estimation method, which closely combines the connections between adjacent segments in the sensor fusion algorithm to obtain drift-free joint kinematics. It can eliminate joint kinematic drift using the common information measured by sensor accelerometers and gyroscopes on the connected segments.

Wearable accelerometers can be used in recognition activities to improve the quality of patient care [34], estimate metabolic energy expenditure during exercise [35,36], measure gait parameters [37], fall detection [38] and detect activities of daily living [39–41]. The main reasons for the widespread adoption of accelerometers are their small size, relatively low cost, and ease of integration with existing sensor network platforms. Moreover, they can show the changes of human motion on motion frequency, intensity and tilt, which is very important for evaluating static posture characteristics. For example, Lu et al. [42] developed a wearable gesture sensing device embedded with three-axis accelerometer and four surface EMG signals to realize real-time interaction of gestures. Lee et al. [43] used inertial motion sensors on gloves to calculate the direction of the steering wheel to evaluate the driver's pressure level. Wang et al. [44] used gait classification to collect foot movement data through accelerometers and then identified gait based on a decision tree classification method. In [45], a calibration method for accelerometer and magnetometer was proposed to deal with ubiquitous orthogonal error and magnetic disturbance. And proportional integral controller based complementary filter and error correction of gait parameters have been defined with a multi-sensor data fusion algorithm.

Gyroscope is helpful to estimate the direction of motion from different angles, which facilitates the differentiation of activity categories and helps to classify activities as zero displacement, transitional displacement or strong displacement activities. Zero displacement activities refer to those activities that do not involve the movement of body parts, such as sleeping, sitting or standing. Transitional activities are those that facilitate the transform from one activity to another, such as from sitting to standing, or from sitting to lying. Intense replacement activities involve walking, running, stairs, running and other physical activities [46].

2.1.2. Physiological sensors

In addition to inertial sensors, physiological sensors such as electroencephalogram (EEG) [47,48], electrocardiogram (ECG) [49,50], galvanic skin response (GSR), thermal sensors [51], and EMG sensors are also used for human activity recognition. Hand movement recognition based on EMG measurement has been widely used in computer and automatic video games, robotic exoskeleton, surgical equipment and power prostheses [52,53]. Zhang et al. [54] proposed a hybrid ECG feature extraction method and a parallel ECG pattern recognition framework to improve the recognition efficiency. Rabin et al. [55] used the non-linear dimensionality reduction method and K neighbor algorithm for data fusion to classify hand movement. Aygun et al. [56] described a new method for robust estimation of physiological parameters, such as interbeatiac interval (IBI) and heart rate variability (HRV) from cardiac signals captured by wearable sensors. In this study, feature extraction technology was used to extract target morphological features candidates from different physiological signals such as R-peaks of ECG or systolic peaks, maximum slopes, and onsets of PPG. In addition, the fusion technology was introduced to take advantage of the different morphological features of single-channel physiological signal morphology. Xiong et al. [57] designed a force display system based on wearable arm posture sensors and electromechanical sensors for upper limb rehabilitation patients to train the end of the arm to apply precise force. In the experiment, a dedicated experimental platform was established to obtain the data of arm end force, arm posture and EMG signal, and generalized regression neural network was introduced as a sensor fusion method to fit the arm end force. The verification in the test process shows that the generalized regression neural network after training is accurate and feasible to predict the arm end force.

Table 1
Related work of reviews.

Theme	Refs.	Publisher	Counts of citations	Highlighted	Year
Multi-sensor	[20]	Elsevier	1696	Data fusion methods given from different data types	2013
	[14]	Elsevier	620	Data-level, feature-level and decision-level fusion in body sensor networks	2016
	[21]	Elsevier	692	Multimodal fusion in affective computing	2017
	[22]	IEEE	146	Distributed fusion estimation algorithms for multi-sensor networked systems	2017
	[5]	CRC	735	Handbook of multisensor data fusion	2017
	[7]	Crossmark	273	Information fusion with both vision and inertial sensors	2017
	[15]	Elsevier	156	Data fusion and multi-classifier systems for human activity	2019
	[4]	IEEE	42	Sensor locations, requirements and challenges	2019
	[17]	Elsevier	68	Emotion recognition with multi-modal data and machine learning	2020
	[23]	ACM	69	Classification of sensors and the challenges of multi-sensor fusion	2021
Deep leaning	[23]	Elsevier	359	Deep learning for human activity recognition	2018
	[24]	Wiley	121	Machine learning and data mining for activity recognition	2018
	[25]	Elsevier	945	Sensor modality, deep model, and application	2019
	[26]	Springer	37	CNN architectures, challenges and medical applications	2021
	[27]	IEEE	6	Wearable Sensors Using Deep Learning Techniques	2021
Applications of HAR	[28]	MDPI	87	Overview of research on fall detection	2018
	[16]	Elsevier	24	Single-user and multi-user activity recognition	2020
	[18]	IEEE	14	Sensor technologies for fall detection systems	2020
	[29]	IEEE	7	Pedestrian dead reckoning with wearable sensors	2020
	[30]	IEEE	21	Users security certification	2020

2.1.3. Pressure sensors

Pressure sensors include force sensing resistors (FSR) [58–60], bending sensors [61], barometric pressure sensors [62], and textile-based capacitive pressure sensors [63]. If the maximum sensing pressure of the sensor exceeds 20 kpa, it can be mounted on the knee joint to monitor human movements such as squatting. If the detection limit of the sensor is as low as about 100 pa, it can be used to detect small physiological activities such as throat vibrations [64,65]. Pressure sensors can also be used as eye movement detectors, such as ultrathin piezoelectric sensors based on piezoelectric nanowires mounted on the user's eyelids to measure eye movements as the eye moves to the left or right at different speeds [66].

2.1.4. Vision wearable sensors

With advances in integrated circuit technology, cameras have been reduced in size and can be worn on the body as wearable vision sensors (WVSs). WVSs have a wide range of applications and are increasingly used in areas such as motion monitoring [67], tele-surgery [68], and telepresence [69].

2.1.5. Flexible sensors

In addition, Most current systems have complex and rigid sensor structures (typically measuring range less than 5% strain), computationally complex data processing and inadequate motion information acquisition (unidirectional deformation only) that can limit their practical application. Compared with rigid sensing systems, flexible sensing systems have greater advantages in terms of power consumption, dynamics, and sensitivity [70,71]. These include:

- Thinner, with high flexibility and bendability, reducing the non-transient effect of its rigid counterpart.
- Better impact resistance than non-flexible sensors.
- Lower cost for mass production and ease of use.
- Substrates used to manufacture flexible sensors are organic, reducing the possibility of thermal damage.

Based on these advantages, flexible sensors are suitable for wearable systems in different applications and have been shown to be a good method for measuring human and robotic motion performance [72–74]. Volkinburg et al. [75] reported a wearable gesture controller fabricated using the sensing capability of a flexible thin-film piezoelectric polymer polyvinylidene fluoride (PVDF). Lu et al. [76] proposed a flexible high-sensitivity humidity sensor and array, which is made by anchoring multilayer graphene (MG) on electrospun polyamide (PA) 66. The sensor detects asthma in a non-contact manner by monitoring respiratory rate in real time and a remote alarm system and provides a non-contact drug delivery interface for bedridden patients. In [77], A pair of gloves, i.e., a sensory glove and a motor glove, was designed and fabricated with a soft, flexible material, providing greater comfort and safety than conventional rigid rehabilitation devices. And machine learning is employed to recognize the gestures from the sensory glove with an accuracy of 93.32%. Although this paper does not introduce too much about flexible sensors alone, and their applications are integrated in the whole paper. There is no doubt that the application of flexible wearable sensors in the field of information fusion will be a very valuable application topic in the future.

2.2. Smart wearable mobile devices

2.2.1. Hand wearable smart devices

With the development of smart phone technology, smart phones are not only communication devices, but also powerful sensor platforms. A smartphone contains many sensors: such as Wi-Fi, Bluetooth, accelerometers, magnetometers, barometers, gyroscopes, proximity sensors, temperature and humidity sensors, ambient light sensors, cameras, and microphones. Since smart phones are widely used in our daily lives and are not interfere with users, they can be good candidates for human activity recognition [78].

Normally users keep smartphones in their shirt pocket, pant pocket or keep it in hand. As accelerometers handle axis-based motion sensing and gyroscope determines the orientation, the position of smartphone needs to be considered while taking data. For different subjects, the positioning and placement of smartphones can be quite diverse because they may have very different dresses, body sizes, and behaviors. To address these issues, a coordinate transformation and principal component analysis (CT-PCA) scheme and an online update algorithm based on an online support vector machine (OSVM) are proposed in [79] to eliminate the effects of orientation, location, and subject variation. Yean et al. [80] proposed a simple feedback loop complementary filter (CFF) to reduce the error caused by the integration of the gyroscope's data in the orientation estimation. Next, they proposed a new and better orientation estimation algorithm which combines quaternion-based kalman filter with corrector estimates using gradient descent (KFGD). The performance of CFF and KFGD in two early rehabilitation training was evaluated. The results show that CFF can achieve fast motion tracking, and the feedback loop can correct the errors caused by gyroscope data integration.

Smart watch is also a powerful multi-sensor device. The eWatch sensor platform used in [81] contains four types of sensors: dual-axis accelerometer, light sensor, temperature sensor and microphone. Recently, Facebook Reality Lab designed a magical wristband device with a built-in electromyography sensor to receive motor nerve signals transmitted from the spinal cord to the wrist and fingers, and convert them into digital commands for controlling the device. The data from the Empatica E4 wristband device consists of several sensors used in combination with sensor fusion methods and machine learning (ML) techniques to be able to determine the physical state of the subject (such as rest, daily life, running, cycling and resistance training) and estimate energy expenditure and physical activity in progress [82]. Experiments have proved that using multiple wristband sensors (skin conductivity and temperature, accelerometer and blood volume pulse) and integrating sensor data can improve the classification and energy of the body state compared to using a single wristband sensor (for example, accelerometer) The accuracy of consumption estimation is used to obtain more reliable physical activity information for treatment decision-making [83,84].

Yuan et al. [85] introduced a gesture recognition framework based on depth feature fusion network of wearable sensors: Firstly, a novel data glove is designed, which integrates two armbands and three-dimensional flexible sensors, and can capture the movement of fine particles from the whole arm and all finger joints. Secondly, an improved deep feature fusion network is used to track long-term dependent complex gestures. Meanwhile, a convolutional neural networks (CNN) based feature fusion strategy is used to fuse data from multiple sensors. In addition, smart gloves can be used for language translation in sign language, smart fitness, etc. [86,87].

2.2.2. Head wearable smart devices

Wearable smart glasses consist of a processor, a miniature projector, sensors, a camera, a control device and a storage device. Smart glasses can be used for a variety of applications, such as medical assistance, fire rescue, travel assistance, etc.

Nguyen et al. [88] proposed a two-channel EEG in the form of earplugs, which showed a stable sleep stage classification with an average accuracy of over 95%. Athavipach et al. [89] designed an in-ear EEG wearable device, using support vector machines to realize the classification of basic emotions. The feasibility is verified by comparing with scalp EEG data (in-ear EEG signals are highly correlated with nearby T7 and T8 scalp signals). Therefore, in-ear EEG can be considered as a substitute for scalp EEG near the ear.

2.2.3. Torso wearable smart devices

Yang et al. [90] integrated inertial measurement units and flexible membrane compression sensors into comfortable sports pants and two foot insoles respectively, and the combined dataset obtained from them can greatly improve the classification performance. Through k-nearest neighbor (KNN) algorithm, the recognition rate of nine motion modes (standing, sitting, running, walking, continuous squatting, up and down stairs and up and down mountains) can reach 99.96%. Ataei et al. [91] introduced the effect of a designed wearable lift-assist vest (WLAV) in the reduction of erector spinae muscle forces during symmetric squat lifting tasks. The device can also be used for back muscle rehabilitation, prevention of back diseases, clinical research, etc. Pino et al. [92] presented a wearable shirt fitted with dry electrodes to measure EMG in upper limbs during physical exercise. The hardware acquires signals from 6 channels and transmits wirelessly to a PC via Bluetooth. The system can monitor EMG activity in real time, providing valuable information for professional and amateur athletes and their sports coaches to improve training protocols.

Fig. 2. The main application of wearable devices in information fusion.

2.3. The main applications of wearable devices in information fusion

In recent years, wearable sensing technology has been widely used in smart video surveillance [93–95], gait monitoring in elderly people, Parkinson's disease or stroke patients [96–99], fall detection in patients with benign paroxysmal positional vertigo [100,101], kicking monitoring in children with Down syndrome [102], bilirubin level monitoring in neonatal jaundice [103], fatigue monitoring in medical personnel, drivers [104–106], and wandering behavior monitoring in patients suffering from Alzheimer's disease (AD) and AD related dementias (ADRD) [107]. Based on information fusion technology, we conclude that wearable sensors are mainly used in the following applications (See Fig. 2).

2.3.1. Military

Military applications are the source of the birth of multi-sensor information fusion technology. With the development of the Internet of Battlefield Things (IoBT), soldiers have become the key nodes for information gathering and resource control on the battlefield. Heterogeneous sensors integrated into soldiers' equipment can provide multi-dimensional battlefield information for the command center. At the same time, as a group of interdependent and interconnected entities, soldiers use IoBT equipment to continuously communicate, coordinate, plan and execute tasks. Therefore, the development of multifunctional military wearable devices has become a trend.

Although the densely deployed wearable sensors provide a platform for comprehensive monitoring of soldiers' status, the wearable technology based on multi-source fusion lacks a general research system and cannot highlight the advantages of heterogeneous sensor networks and information fusion. Shi et al. [108] proposed a multi-level fusion framework based on soldier body sensor network, and described the deployment model of heterogeneous sensor network. The framework covers a variety of information types on a single node, including behavior, physiology, emotion, fatigue, environment, and location, so that soldiers can obtain accurate evidence, decision-making capabilities, and information flexibility under resource constraints. In addition, according to the framework structure, the problems and solutions of each unit are systematically discussed, and the research direction is determined for the development of military wearable devices. Ruotsalainen et al. [109] discussed the statistical model of measurement error of

inertial sensor and vision based heading and translation measurement. Particle filter (PF) method was used to integrate the measurement results of pedestrian motion with non-Gaussian error characteristics, and its horizontal positioning effect was better than that of Kalman filter.

2.3.2. Emotional calculation

Emotion is a more complex and stable physiological evaluation and experience of human attitude in physiology, which is reflected in physiological reactions including seven basic emotions: joy, anger, sadness, thought, grief, fear and surprise. The main contents of emotion computing include: analyzing the mechanism of emotion, mainly the determination of emotional state and the relationship with physiology and behavior; using a variety of sensors to obtain information on the behavioral characteristics and physiological changes in the current emotional state of human, such as voice signals, facial expressions, body postures and other body language, as well as pulse, skin electricity, brain electricity and other physiological indicators; through the analysis and processing of emotion signal, the emotion model is constructed to quantify the emotion, so that the robot has the ability to perceive, recognize and understand the human emotion state [110]. In health care, intelligent systems with emotional interaction capabilities can capture users' physiological signals related to emotional changes through smart wearable devices in a timely manner, and when monitoring users' large emotional fluctuations, the system can regulate users' emotions in a timely manner to avoid health hazards or make health care suggestions. In distance education, the application of computational emotion can improve learners' interest and learning efficiency and optimize the function of computer-assisted human learning. In e-commerce, the system can track the user's eye trajectory, focus and other parameters when browsing design solutions by eye-tracking, analyzing the correlation between the parameters and customer attention, and record the customer's interest in products and automatically analyze their preferences.

Previous studies have shown that the emotion recognition systems based on multi-source data fusion can be obtain higher accuracy and more reliable results than a single sensor, even if the fast decision-level fusion method is adopted [111]. Dashtipour et al. [112] proposed a new context-aware multimodal sentiment analysis framework with decision-level (late) and feature-level (early) fusion methods to integrate sentiment cross-modal information. Physiological parameters used in recent studies include EEG, ECG [113,114], EMG [115], photoelectron plethysmography (PPG) [116], body temperature [117], and facial features. Hssayeni et al. [118] developed a multi-modal physiological data fusion framework by using deep CNN to collect motion and physiological signals collected from wearable devices (such as respiration (RESP), electrocardiogram (ECG), electromyogram (EMG) and electrodermal activity (EDA)) to estimate scores of positive and negative effects, and in the process investigated two different types (gradient augmentation trees and convolutional neural networks) of data fusion methods and considered the effects of different types of physiological signals in the estimation. Khezri et al. [119] added the electrooculography in the data fusion process. Mohino-Herranz et al. [120] analyzed the electrocardiogram and chest impedance record of the heart and respiratory activity monitored by wearable sensors, and can distinguish three different emotions: neutral emotion, sadness and disgust.

2.3.3. Sports activities

The rapid change of motion posture will obscure the camera position, and the use of wearable sensing devices solves this problem. Therefore, wearable sensors are essential for sports activity recognition.

In the field of motion recognition, the main advantage of multi-sensor fusion methods compared to single-sensor and multi-sensor fusion methods is that they can provide more information about limb motion and thus improve the recognition ability of subsequent recognition algorithms. Wang et al. [121] proposed a feature subset selection framework of sensor combination for motion phase segmentation, which combines feature selection algorithm with subsequent classifiers to determine the best combination of sensor and feature subset according to the performance of training model. The experimental results show that the proposed framework can effectively select a certain number of sensors without human intervention, and the influence of the number of sensors on the recognition rate of the classifier is less than 1.5%, which is effectively applied to the phase segmentation of gait and breaststroke.

Wang et al. [122] proposed a portable and quantifiable deadlift evaluation system, which employs a human body sensor network (BSN) including inertial and surface electromyography (sEMG) information to reconstruct and evaluate 3-D posture, so as to compare the joint angle and muscle state of coaches. A two-step extended Kalman filter (EKF) for attitude updating is proposed in data fusion, and the accuracy of the algorithm is verified by optitrack system. Wilk et al. [123] constructed a computationally efficient three-dimensional (3-D) attitude detection wearable device by using multimodal sensor fusion algorithm and the complementarity of vision and inertial sensor modes. By fusing the information of the two external reference points captured by the camera with the azimuth captured by the inertial motion sensor, the three-dimensional attitude can be determined, which can be well applied in tracking barbell squatting movements. Wang et al. [124] used multi-sensor data fusion algorithm to estimate the posture of swimmers, and reconstructed the swimming pose combined with the human biomechanical model. Liu et al. [125] used multiple micro inertial sensor nodes to collect the movements of kayakers. They used the gradient descent method to fuse the data, obtained the pose information of the canoe after sensor calibration, and then reconstructed the action motion of the canoe. Fang et al. [126] proposed a new method to reconstruct the position and speed in the background of ski jump, which integrates the sensor data including inertial measurement unit and GPS recorder. Tora et al. [127] used deep learning (DL) combined with the context of ice hockey game activities to aggregate multi-user data and classify group activities in ice hockey games. Kautz et al. [128] designed a deep convolutional neural network based on wearable sensor activity classification to monitor beach volleyball players and avoid injury caused by excessive exercise. Correct riding posture is crucial for good performance in equitation, therefore, an equestrian analysis method based on human sensor networks is proposed in [129]. First, using inertial sensor nodes to collect raw sensor data corresponding to the different gaits of professional riders and beginners. Then, after reducing the error after the initial sensor alignment, a motion reconstruction method based on a gradient descent approach is used to fuse the sensor data to continuously update the rider's posture information.

2.3.4. Medical care

Sharp and Hockfield noted that wearable smart devices can monitor health conditions, alert patients and physicians to early health problems, and help in disease prevention. At the same time, algorithms for data-driven medical decision-making can help doctors use the best available resources [130]. The effects of information convergence on traditional medicine are presented in Fig. 3.

On the one hand, the increasing pressure of human existence in modern society has led to a trend of younger development of chronic diseases such as hypertension and heart disease, and their number is increasing year by year. Therefore, out-of-hospital monitoring of patients suffering from chronic diseases to obtain parameters of physiological and psychological changes in real life is important to assess the effectiveness of home treatment. On the other hand, the probability and frequency of accidental falls in the elderly are high due to the aging of their physiological structure and the decline of their physical functions. In daily life, especially in wet places such as kitchens, bathrooms and toilets, elderly people are prone to slip and fall. Falls may also threaten the lives of elderly people if they suffer from brain hemorrhage, heart disease and other illnesses.

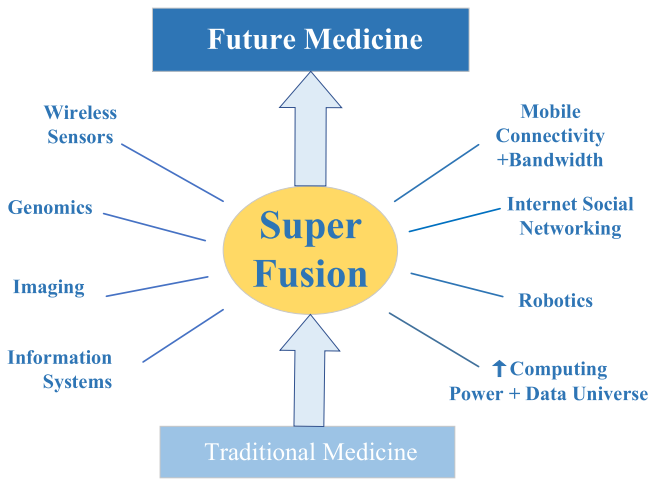

Fig. 3. The effects of information convergence on traditional medicine.

Currently, most fall detection algorithms based on single-sensor data cannot accurately describe the fall status, while fall detection algorithms based on multi-sensor data integration can improve the sensitivity and specificity of prediction. Pan et al. [131] designed a fall detection system based on multi-sensor data fusion and analyzed four stages of falls using data from 100 volunteers with simulated falls and daily activities. The data fusion method it used was able to extract three characteristic parameters of human acceleration and posture change with a sensitivity of 96.67%, which verified the effectiveness of the multi-sensor data fusion algorithm.designed a fall detection system based on multi-sensor data fusion, which used a data fusion method capable of extracting three characteristic parameters of human acceleration and posture change with a sensitivity of 96.67%. Aleman et al. [132] presented a tele-rehabilitation system that uses the motion-tracking sensor of the Kinect, to allow the elderly users natural interaction, combined with a set of external sensors as a form of input. Data fusion techniques were used to integrate these data to detect right shift and monitor the rehabilitation process of the elderly during treatment. A tele-medicine service mode is shown in Fig. 4. Fortino et al. [133] presented a few applications that have been built atop SPINE, an open-source programming framework, such as physical activity recognition, rehabilitation support, physical energy expenditure estimation, and gait analysis.

Li et al. [134] proposed a method to evaluate the therapeutic effect of hip joint based on human body sensor network (BSN) for children with cerebral palsy (CP). In this method, distributed magnetic, angular rate and gravity sensors are used, and the total motion function of CP children is evaluated by multi-sensor data fusion algorithm. The results of comparison with the golden standard optical capture system prove the robustness of the sensor fusion algorithm. Obstructive sleep apnea (OSA) is a chronic sleep disorder. People with OSA rarely realize their condition, but long-term untreated can lead to more serious conditions. Gutta et al. [135] proposed a new OSA detection framework that combines wearable sensor measurement signals with a mathematical model of the cardiopulmonary system, which can simulate physiological changes between different individuals.

Chen et al. [136] proposed a system framework that fuses different sensor measurements to simultaneously track human joints and model upper extremities to provide estimates during occlusion. To fuse the measurements from two or more different sensors, they determine the spatial relationships between sensors and simultaneously locate sensors and track the moving joints. Firstly, the state vector is extended to 3-D space and the directional representation is replaced with quaternions to reduce linearization errors. Secondly, three measurement models (position-based, acceleration-based and orientation-based) are used to take into account the different observations. Finally, the sensor and joint trajectories are estimated and the parameters of the kinematic model are estimated using a Kalman filter algorithm.

2.3.5. Pedestrian navigation

In the past decade, the target localization and tracking technology of objects or individuals has been developing rapidly, which has greatly improved modern life. For example, providing navigation for commuting in daily activities [137], or positioning to save lives in disaster relief or medical emergencies.

The method based on human body positioning applies small wearable sensors, including accelerometers, gyroscopes, compasses, barometers, magnetometers, anemometers, and body cameras. Hou el al. [29] systematically reviewed all researches related to pedestrian dead reckoning (PDR) with wearable sensors. They pointed out that the filtering algorithm integrating multiple observations is very important in the pedestrian dead reckoning system with multiple sensors, and the most commonly used data fusion algorithm is the Kalman filter. In [138], a new pedestrian dead reckoning method for head-mounted sensors was proposed, whose data are collected by sensors placed in glasses and embedded in a mouthguard.

Zhao et al. [139] proposed a new three-dimensional position estimation method, which consists of a sensor device integrated with accelerometer, gyroscope, magnetometer, and barometer. Firstly, the vertical position of the pedestrian is estimated by fusing acceleration and angular velocity and height derived from barometer data. Secondly, the horizontal position of the pedestrian is estimated using a pedestrian heading projection mechanism, and an adaptive multi-peak step model and a multi-sensor fusion-based heading angle estimation method are proposed.

Zhou et al. [140] proposed a walking navigation method based on virtual inertial measurement units (VIMU) and gait feature assistance: the inertial data from different locations of pedestrians' lower limbs were collected synchronously by actual IMUs as training samples, and a visual geometric group long short-term memory (VGG-LSTM) neural network model was used to establish a nonlinear mapping relationship between human foot and leg inertial information, from which a foot virtual inertial measurement unit and a virtual inertial navigation system (VINS) were constructed. Hasan et al. [141] proposed an indoor navigation system to track the location of pedestrians by using the built-in inertial measurement unit (IMU) sensor of the "JINS MEME" smart glasses. Extended Kalman filter was used as a sensor fusion algorithm to fuse accelerometer and gyro data to determine speed and heading. Zhao et al. [142] fused the inertial data with only a fraction of magnetometer data that are accurate and usable, and improved the gradient descent algorithm (GDA) to reduce heading drift.

3. Information fusion for multi-modal wearable devices

In this paper, we divide information fusion of wearable sensors into two methods: multi-modality and multi-location, as shown in Fig. 5.

3.1. Multi-modality sensor fusion

Sensor fusion is an effective method to reduce uncertainty, enhance reliability. The rationale behind this theory is that one source can compensate for errors in another source, thus providing advantages such as improved accuracy and fault resilience. For example, accelerometers by themselves may not be sufficient to provide enough useful information. They need to be combined with other sensors such as microphones [143], gyroscopes [78,81,144], and ECG sensors [79] to provide a more accurate classification of activity. Wu et al. [145] also discussed the limitations of using only accelerometers. While simple and cost-effective, accelerometer-based systems can still generate false alarms even with multiple sensors, especially in environments such as high-speed trains and elevators where acceleration can interfere

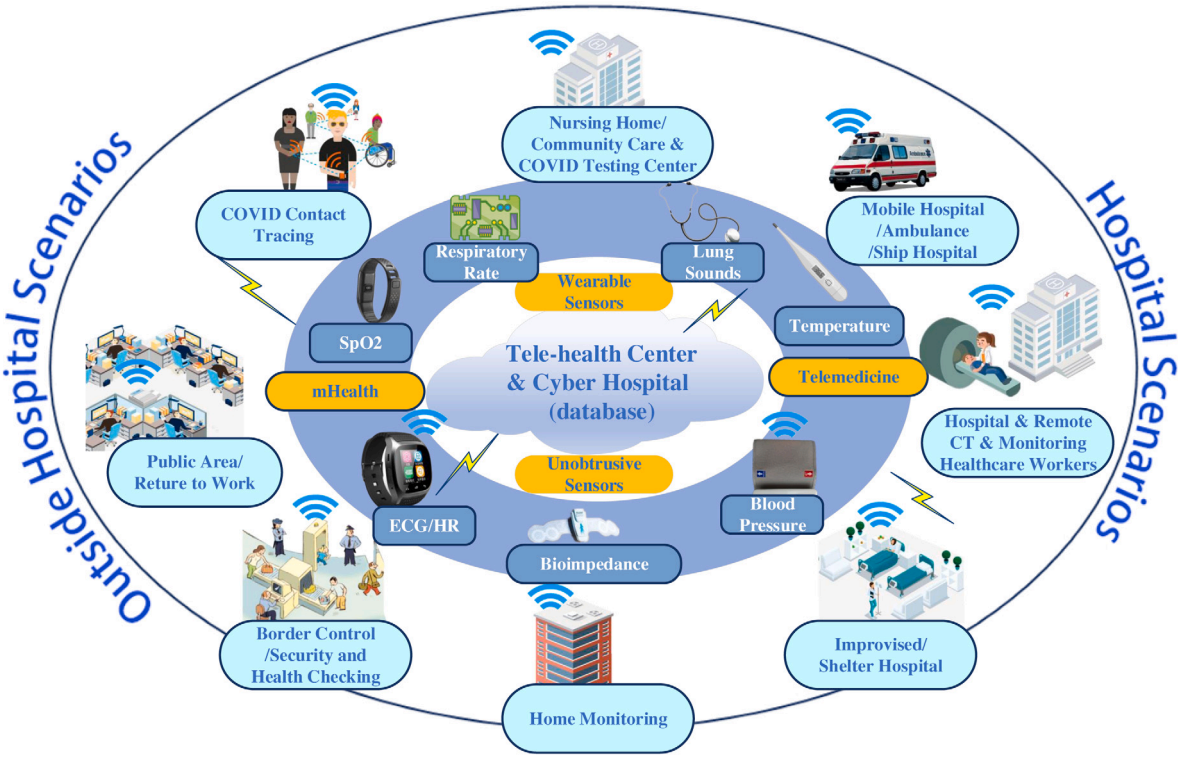

Fig. 4. A tele-medicine service mode.

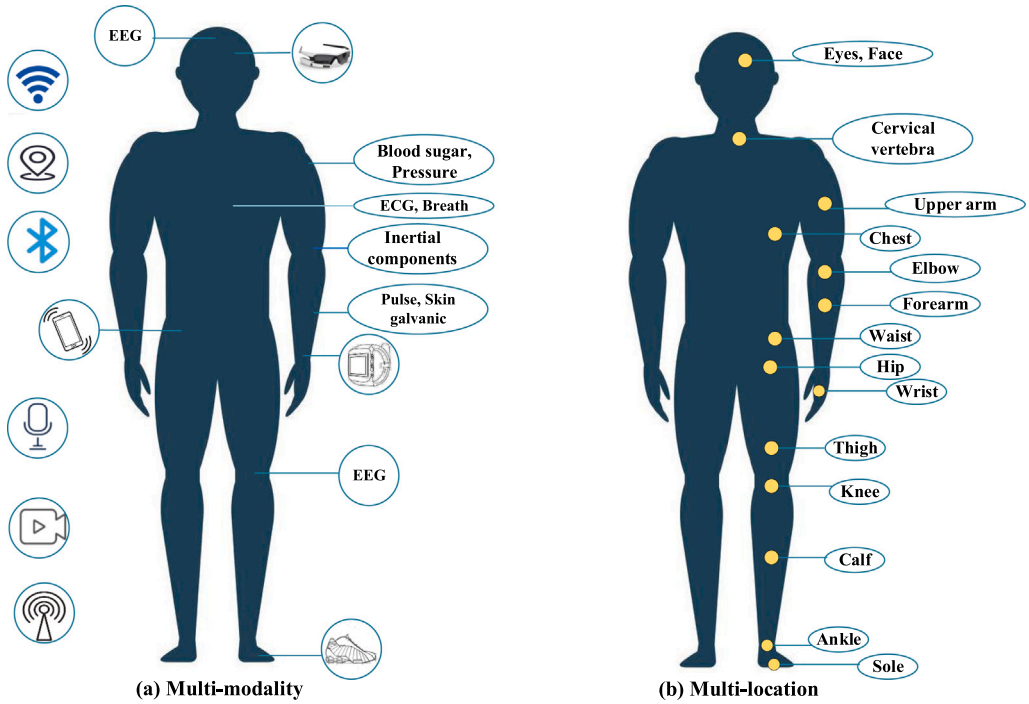

Fig. 5. Wearable sensor information fusion mode.

with the recognition of human activity. The following describes the fusion modes of several heterogeneous sensors. Table 2 summarizes the multi-modality application information based on mobile and wearable sensors.

3.1.1. Environmental and wearable sensors

Environmental sensors include ultrasonic, pressure sensors, temperature sensors, infrared sensors, and radar sensors, etc., and are often

Table 2
Sensor modalities, applications, methods, accuracy and datasets.

Applications	Refs.	Year	Multi-sensor	Methods	Accuracy	Datasets
Multi-resident activity Recognition in a smart home	[146]	2018	Environmental+Motion +Temperature sensors	CNN	95.20%	Cairo
Fall detection	[147]	2017	Radar+Wearable+Kinect	SVM	91.36%	Collection
	[131]	2020	Electrochemical+Pressure+ Stress+Magnetic field sensors	SVM	96.67%	Collection Collection
	[148]	2020	Inertial sensors	Linear predictive+RF	97%	SCUT-NAA
Chinese sign language recognition	[149]	2019	sEMG+ACC+GYR	DBN	95.10%	Collection Collection
Hand gesture recognition	[85]	2020	ACC+GYR+Bending	KNN SVM LSTM CNN CNN+LSTM	93.06% 94% 99.81% 99.82% 99.93%	ASL+CSL ASL+CSL ASL+CSL ASL+CSL ASL+CSL
	[150]	2020	SEMG+Inertial sensors	Dictionary learning	97.40%	CapgMyo
Bad posture classification	[150]	2020	ACC+MAG+Miniature magnets	DNN CNN	88.10% 88.70%	Collection Collection
Physical states classifier	[83]	2020	Skin temperature+ACC+GSR+Blood volume pulse+PPG	SVM DT KNN LSTM	85.34% 88.57% 92.35% 94.80%	Collection Collection Collection Collection
Violin bowing actions recognition	[151]	2020	Depth frames+ACC+GYR	CNN+LSTM	80.90%	Collection+Data Augmentation
Swimming phase segmentation (Arm,Leg)	[152]	2020	ACC+GYR+MAG	DT NB KNN LR SVM	93.66%,96.47% 94.89%,96.52% 95.13%,96.67% 95.47%,96.85% 96.39%,97.67%	Collection Collection Collection Collection Collection
Detecting drowsiness	[153]	2021	EEG+EMG	CNN	94.87%	Pz-Oz
Paddle Stroke Analysis for Kayakers	[153]	2021	ACC+GYR+MAG	LR DT RF SVM KNN	97.89% 98.27% 98.56% 98.56% 98.98%	Collection Collection Collection Collection Collection
Gait analysis	[154]	2021	Inertial sensors	HMM+MAP	91.88%	Collection
	[155]	2021	Inertial sensors	LSTM	93.14%	whuGAIT
HAR	[156]	2018	Smartphone inertial sensors	ANN SVM DBN	89.06% 94.12% 95.85%	Collection Collection Collection
	[157]	2019	ACC+GYR	LSTM	93.13%	UCI
	[158]	2020	ACC+GYR+MAG	ANN	93%	Collection
	[159]	2020	Smartphone inertial sensors	LSTM CNN-LSTM LSTM CNN-LSTM	91.28% 92% 96.23% 99%	UCI UCI iSPL iSPL
	[160]	2021	Smartphone inertial sensors	Vanilla LSTM 2-stacked LSTM 3-stacked LSTM CNN-LSTM 4-layer CNN-LSTM	96.54% 97.32% 96.59% 98.49% 99.39%	UCI UCI UCI UCI UCI

used to observe environmental changes to detect human activities in specific environments (such as homes and workplaces).

Tan et al. [146] used environmental sensors (PIR and temperature sensors) data to generate a deep convolutional neural network classifier based on RGB activity images to achieve clear recognition of multi-resident activities. Pham et al. [161] used motion information from distributed environmental sensors and body activity information from wearable sensors to localize inhabitants in an indoor environment. Li et al. [147] solved the performance limitation of using a single sensor by fusing the experimental data collected by different sensors such as three-axis accelerometer, micro-Doppler radar and depth camera, especially in the classification of similar activities. Preliminary results confirm that combining information from heterogeneous sensors can improve the overall performance of the system. The classification accuracy obtained by this fusion method is 11.2% higher than that of using radar alone, and 16.9% higher than that of using accelerometers. Anacleto et al. [162] proposed a pedestrian inertial navigation system that includes force sensors combined with accelerometer data to have a better detection of the stance phase of the human gait cycle, leading to an improvement in position estimation. In addition to sensor fusion, an information fusion architecture is proposed for learning pedestrian gait behavior based on information from GPS and several inertial units placed on the pedestrian to correct inertial sensor errors in real time, leading to improved position estimation. Wu et al. proposed a multi-modality wireless sensing approach for inferring mobile behaviors from Wi-Fi signals, GPS and smartphone accelerometer measurements [163].

3.1.2. Visual/auditory and wearable sensors

(1) Image Sensor:

In an image-sensor-based system, the subject is monitored in real time by a camera, and various algorithms can be used for pose recognition [164,165].

a. RGB/Camera:

Ozean et al. [67] showed that fusion of camera and accelerometer data not only improves the detection rate, but also reduces the number of false alarms compared to systems based only on accelerometers or cameras only. Chen et al. [7] studied the fusion of video and inertial sensors for human activity recognition. The Wearable Mobility Monitoring System (WMMS) proposed in [166] became a useful tool for rehabilitation decision making. It uses an integrated three-axis accelerometer, GPS, camera, and timer to identify mobility changes-of-state (CoS) between static activities, walking-related activities, elevator rides, kitchen work, and cooking.

b. Depth Camera:

There are three types of commonly used depth cameras: stereo cameras (measured by a tripod), time-of-plane-based cameras, and structured-light-based cameras. Compared with the image generated by the traditional camera, the depth image generated by the depth camera is not sensitive to the change of illumination, and achieves high recognition accuracy, especially in the recognition of indoor human movements. Since depth images provide a rich representation of global (or whole-body) motion conditions, and inertial signals capture local motion characteristics, the fusion of depth images and inertial sensors makes the recognition accuracy higher.

Hwang et al. [167] obtained the motion parameters of the object by using inertial sensors, and obtained the same motion information as the actual motion by using the information about the actual joint length and initial direction of the object extracted from the depth camera, so as to analyze and correct the golf swing posture.

Sun et al. [151] proposed a recognition system that uses a depth camera and an inertial sensor worn on the violinist to perform information fusion, which can accurately identify different bowing movements in classical violin performance: Firstly, the dataset of violin bow movement is constructed by using the data obtained from depth camera and inertial sensors. Secondly, the depth frame data is enhanced by rotation in 3-D world coordinate system, and the inertial sensor data is enhanced by yaw, pitch and roll angle transformation. Finally, the bow action classifier is trained in different ways to make up for the advantages and disadvantages of various methods based on deep learning and decision level fusion.

Chen et al. [168] considered a depth motion map (DMM) extracted from a depth camera as a way to obtain action-related motion information. In addition, four statistical features (mean, variance, standard deviation, and root mean square) were extracted from the tri-axial acceleration and tri-axial angular velocity signals of the wearable inertial sensor.

(2) Acoustic Sensor:

Witchit [169] proposed a multisensory data transfer architecture that combines video cameras, infrared, acoustic and pressure sensors and uses a fuzzy inference machine for information fusion. The integration of sensors improves the accuracy and robustness to recognize different activities (e.g., bending, lying, reading, sitting, and eating).

FamilyLog is the first practical system for recording family mealtime activities with smartphones and smartwatches. It uses the smartphone/smartwatch's built-in accelerometer and microphone to detect mealtime activities that are closely related to family health, including meal times, durations, the number of people participating in family meals, and conversations and TV viewing during meals. It classifies family members' behavior using a novel conditional random domain-based sensor fusion technique after feature extraction by built-in sensors [170].

Arabaci et al. [171] used visual, audio and wearable sensor information to extract different feature sets and used two multi-core learning (MKL) techniques (MKBoost [172] and SimpleMKL [173]) for feature fusion.

3.1.3. Physiological signals and wearable sensors

Physiological signals include electroencephalographic signals, respiratory signals, cardiac signals, electromyographic signals, and electrical skin responses. Tapia et al. [174] investigated human activity recognition and health status monitoring combining accelerometers and physiological signals, such as the fusion of ECG scanners and accelerometers for monitoring the activity of stroke patients. In addition, the fusion of accelerometers and ECG scanners provides functional means to estimate physical exertion, track health management problems, and prevent diseases, leading to lifestyle improvements [175].

Jia et al. [176] proposed a new scheme to recognize human daily activities by fusing accelerometers and ECG. First, time and frequency domain features are extracted from the raw sensor data. To reduce the computational complexity of the subsequent process, a linear discriminant analysis (LDA) is performed thus will significantly reduce the size of the feature vector. Then, the feature vectors are reduced and classified by a relevance vector machine (RVM). Finally, considering that different sensors and cues will provide complementary information about human activity, individual classification results are fused at the decision level to improve the overall recognition performance. Experimental results show that even recognition accuracies of up to 99.57% can be achieved if the 7-lead data from ECG and accelerometer are fused together.

The fusion of accelerometer and barometric pressure sensors can also play an important role for the detection and prevention of child-related unintentional injuries. In [177] using only wearable tri-axial accelerometer sensors, the average overall accuracy of SVM and decision tree was 86.2% and 88.3%, respectively, while using wearable three-axis accelerometer and barometric pressure sensors, the average overall accuracy of SVM and DT was 98.43% and 96.3%, respectively.

3.2. Multi-location sensor fusion

A major challenge in human activity recognition is how to identify concurrent activities. Some types of activities occur simultaneously, including brushing teeth while walking, lying down and talking on the phone. In these cases, the activities to be recognized are segmented into lower and higher activities by a hierarchical approach. Multiple acceleration sensors are attached to the chest, left forearm, right thigh, left ankle, and right wrist, leaving the entire body in motion monitoring [178]. Raj et al. [2] classified human activities into five major categories, and for very simple activities (e.g., sleeping), wrist wear sensors typically provide reasonable recognition accuracy rates of over 85%; for lower level activities (e.g., eating, reading), the waist sensor was selected by all classifiers as the one that provided the greatest accuracy and recall rate; for intermediate activities (e.g., walking, doing housework), the chest and wrist sensors provided the best accuracy rates; for advanced activities, they were mainly collected by sensors at the arms and knees, as they measured changes in body posture during walking and running.

When the device is placed on different locations, the inertial sensors may exert different forces, and the distribution of sensor data from different locations is considerably different, even when the user is performing the same activity [179]. Sun et al. [180] considered the correlation between sensor data from different locations based on human behavioral characteristics. They proposed that the combination of sensor information from different locations can better reflect the human motion characteristics and effectively tap the potential information in the available data, which in turn improves the behavioral recognition rate. Janidarmian et al. [181] performed a comprehensive human activity recognition using accelerometers placed on 10 major parts of the body, such as upper arms, ankles, and chest. Nweke et al. [182] used accelerometers and gyroscopes for human activity recognition, where the best results were obtained with sensors placed on the chest and wrist.

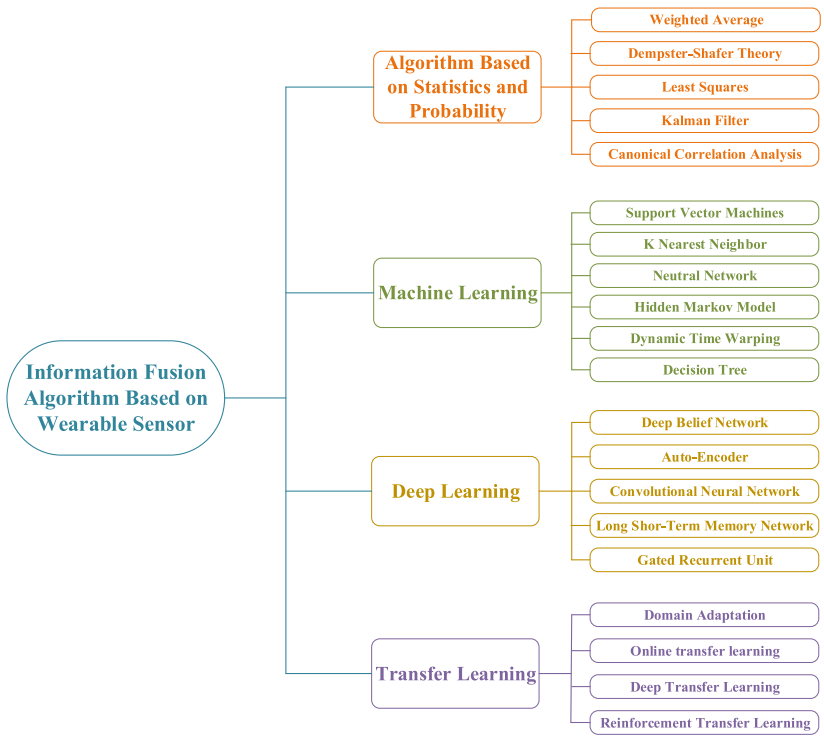

Fig. 6. Multi-mode sensor information fusion algorithm.

The placement of the ankle joint sensor can accurately identify activities involving leg movements, such as running, walking, and standing, while the placement of the wrist joint sensor can fully identify activities such as reading, smoking, and eating [183]. Ugulino et al. [184] suggested the use of multiple accelerometers to distinguish between walking, sitting, standing, going down the stairs, lying in bed, sitting on a chair, walking forward, cycling, jogging and jumping. The accelerometer is placed in different positions such as waist, thigh, arm, wrist and ankle to achieve relative performance improvement. Of course, this requires the wearing of a large number of sensors, which adds additional burden and disruption to geriatric patients, even as the use of smartphones presents other challenging issues.

4. Multi-mode sensor information fusion algorithms

Feature fusions provide excellent means to combine heterogeneous sensor data. This section will focus on manual feature extraction algorithms based on probability and statistics and machine learning, as well as deep learning fusion algorithms for automatic feature representation. Our presentation of fusion algorithm is organized shown in Fig. 6. Table 2 summarizes the recent applications domain in mobile and wearable sensors based human activity recognition, the accuracy of each methods. And Table 3 compares the advantages and disadvantages of common methods of HAR.

4.1. Method based on statistics and probability

4.1.1. Weighted average and least squares method

In the process of data acquisition, a large number of errors and noises may be introduced due to time lag, incorrect sensor position, positioning and sensor failure. The weighted average and least squares methods can provide technology for simultaneous interpreting sensor data, or the average of error sensor readings placed at different sensor locations [189,190].

Tryon et al. [191] found that using weighted average fusion method for EEG and EMG signals and combining velocity information into the model can provide the best performance. Zhou et al. [192] measured the spatiotemporal gait characteristics of different neurological patients based on wearable devices and applied partial least squares discriminant analysis (PLS-DA) to classify fallers and non-fallers. Arunkumar et al. [193] proposed a new de-noising technique using the hierarchical structure of cascade and parallel combinations of two different pairs of adaptive filters. The first pair combines normalized least mean squares (NLMS) and recursive least squares (RLS) adaptive filters and the second pair combines recursive least squares and least mean squares (LMS) adaptive filters.

4.1.2. Kalman filter and extended Kalman filter

Kalman filtering, a predictive update fusion method proposed by Kalman Rudolph in 1960 [194,195], is an effective method for processing time-interval signal sequences. This method minimizes the posteriori estimated covariance by calculating the relative confidence and estimated covariance between the estimates of past observations and the current observations given a sequence of observations from different sensors. Tunca et al. [196] proposed a movement-based pathological gait analysis method with accelerometer and gyroscope sensors mounted on the foot and using the Kalman filter algorithm. This approach is computationally efficient and can provide techniques for fusing accelerometer and gyroscope data to provide better estimates for linear filtered systems and static structure analysis. Yao et al. [197] used Kalman filtering in the feature space to increase specificity by 17.0% and accurately and reliably detect tremor episodes in Parkinson's disease (PD) with a reduced false alarm rate.

Kalman filtering algorithm provides the estimated value of drift corrected signal by using the latest set of measurements and comparing them with the predicted value. However, Kalman filter is limited to linear and Gaussian transform, which is sometimes impractical in real-time applications. In addition to the conventional linear model of Kalman filter, nonlinear model or extended Kalman filter has been used to solve the nonlinear problem of signal drift in inertial sensors [198,199]. In recent years, extended Kalman filter (EKF) [200,201], multiplicative extended Kalman filter (MEKF) [202,203] and unscented Kalman filter (UKF) [204] have been developed to deal with

Table 3
Strengths and weaknesses of information fusion algorithms.

Methods	Strengths	Weaknesses
HMM	Given a model and a particular output sequence, find the sequence of states most likely to produce this output sequence	HMM models are memoryless and cannot take advantage of contextual information. Because it only related to the previous state, if we want to take advantage of more known information, we must build a high-level HMM model [154]
EKF	Kalman filter is proved to be the best estimate in linear problems	It can only estimate linear process model and measurement model accurately
CNN	CNNs are particularly advantageous with regard to local dependency and scale invariance [185]	Require large dataset and high number of hyper-parameter tuning to achieve optimal features [186]
RNN	RNNs are able to learn arbitrary nonlinear dynamical mappings, such as those commonly found in nonlinear time series prediction [187]	Large parameter update is challenging for real-time activity predictions [186]
LSTM	The advantage of LSTM networks over other types of RNN is that the dependency of the current on the previous hidden state is designed in such a way that the LSTM obtains the ability to keep parts of its hidden state over a larger number of time steps [187]	More parameters, easy to fit
GRU	The modified GRU module in RNN is similar to Long short-term memory (LSTM), but it has a simpler structure. In particular, it takes less time to learn due to fewer parameters than LSTM. Also, it is possible to learn with a small number of data [188]	The LSTM is strictly stronger than the GRU because it can easily count indefinitely, whereas the GRU cannot
SAE	Produce more linearly separable features [186]	High resource consumption and low rate of learning
DBF	Unsupervised training with unlabeled sensor streams which is naturally available through cyber–physical systems and Internet of Things and initialization prevent	Mobile and wearable sensor on-board training of the network is computationally complex due to extensive parameters initialization process

the ranging problem from sensor or attitude, attitude instability to sensor position.

Liu et al. [205] proposed a systematic method to evaluate the training performance of canoeists based on multi-sensor fusion technology: Firstly, the motion information of the kayak is collected through the micro inertial sensor nodes attached to the human body. Secondly, EKF is used to update human posture by data fusion. After sensor calibration, the movement of canoeing athletes is reconstructed through rigid body model, and then quantitative kinematics analysis is carried out based on joint angle. Finally, using machine learning algorithm, the stroke cycle is divided into different stages, including entry, pull, exit and recovery. Li et al. [206] used an extended Kalman filter for fusing sensor signals to measure the motion of their body parts, while verifying the accuracy of the multi-sensor fusion algorithm with the optical system OptiTrack and reconstructing the motion of children with motor dysfunction. Qiu et al. [207] used an extended Kalman filter to solve the self contained sensor-based pedestrian heading projection (PDR) faced with sensor installation error and path integration error due to sensor variance.

The advantage of the multiplicative extended Kalman filter over the extended Kalman filter is that the error between the estimated direction and the true direction is much smaller than the error in the actual direction, ultimately leading to smaller linearization errors. Moreover, the MEKF requires lower computational cost [208], which makes it more attractive for real-time applications. Vargas-Valencia et al. [209] developed and validated a knee sleeve-based sensor fusion system for estimating knee flexion and extension angles. The system consisted of data merged from two inertial sensing units and a polymer optical fiber (POF) sensor based on intensity variation, estimated using a quaternion-based MEKF. Rezaei et al. [210] used data from two gyroscopes and one stretch sensor to accurately track the three-dimensional motion of the joint, including extension, flexion, and rotation. And the UKF algorithm was used to directly fuse the angular velocity/relative orientation data to estimate the motion orientation of the joint.

4.1.3. Dempster–Shafer evidence theory

Evidence theory was proposed by Dempster in 1967 and later expanded and developed by Shafer [211], so evidence theory is also known as D–S theory. It is a data fusion method that first calculates sensor reliability and then combines specific combination rules [212].

Chen et al. [213] proposed a fusion method for human action recognition based on a depth camera and an inertial sensor. Computationally valid action features are extracted from the depth image provided by the depth camera and the accelerometer signal provided by the inertial body sensor. In feature-level fusion, features generated from two different modal sensors are combined before classification, while in decision-level fusion, Dempster–Shafer theory is used to combine the classification results from two classifiers, each corresponding to one sensor.

4.1.4. Canonical correlation analysis

Canonical correlation analysis (CCA) can learn the complex nonlinear transformation of heterogeneous data patterns by minimizing the total correlation effect, and produce the output representation with high linear correlation. This method allows data to be represented by different patterns, and integrates these patterns to enhance the ability of human activity recognition. At present, canonical correlation analysis has been used to fuse accelerometer and gyroscope data for human activity recognition [110].

Ha et al. [214] proposed a wearable headband and earbud shaped multimodal psychological management system for monitoring electroencephalogram (EEG), blood electroencephalogram (HEG) and heart rate variability (HRV) for accurate mental health monitoring. After acquiring EEG, HEG and HRV, low-pass filtering based on finite impulse response (FIR) filters was applied to suppress noise. Using the features extracted from each modality, different modalities can be combined for maximum correlation using the CCA and temporal kernel canonical correlation analysis (tkCCA) algorithms to induce coupling of the neuro-vascular-vegetative nervous system. Hwang et al. [215] proposed a novel method of life recording using non-visual information with the help of visual information. CCA can be used to calculate weights when extracting visual and non-visual features.

4.2. Machine learning algorithms

Machine learning methods provide a better mechanism for accurate detection activities by realizing effective data preprocessing and segmentation, feature extraction and dimensionality reduction, classification of learning models, and final evaluation of learning algorithms [9]. The main methods include support vector machines [78, 103,150,152,216–221], k-nearest neighbors [150,221,222], dictionary learning [150,223], Naive Bayesian [217], decision trees [221,224], artificial neural networks (ANN) [158,225], hidden Markov models [226,

227], linear discriminant classifiers [228,229], random forests [148,150,215,230], k-means clustering [3] and dynamic time regularization, etc. Guo et al. [223] established a multi-sensor monitoring system for aerobic exercise recognition and proposed a recognition algorithm based on dictionary learning algorithm and sparse representation. Spinsante et al. [231] used decision tree algorithms to integrate motion sensors to evaluate the effectiveness of activity monitoring models to prevent sedentary lifestyles in the workplace.

Dai et al. [232] developed a wearable biosensor network for emotion recognition in daily life. The scheme enables network recognition by collecting and recording EEG, pulse, skin temperature and blood pressure through sensor nodes and transmitting them to a remote network data center. For an unbalanced and noisy dataset that may arise from the significantly different appearance of four emotional states (fear, happiness, boredom and relaxation) and noisy perceptual environments, they propose an unbalanced fuzzy support vector machine (RI-FSVM) classification method to reduce the adverse effects of noisy intra-class samples and inter-class imbalance.

Hidden Markov models (HMM) have been widely used in the fields of speech recognition [233] and human activity recognition [234]. Liu et al. [154] found that performing gait analysis based on HMM models and parametric adaptive model to segment gait phases can obtain high recognition rates. Li et al. [235] proposed an improved HMM-based HAR method based on the assumption that human behavior can be fundamentally recognized by three key poses around the initial, intermediate and final action cycles. In [236], the recognition problem of concurrent activities is explored with the framework of parallel hidden Markov model (PHMM), where two basic HMMs are used to model the upper limb movements and lower limb states, respectively. In addition, there are abstract hidden Markov models (AHMM) [237], hierarchical hidden Markov models (HHMM), and maximum entropy Markov model (MEMM). Nguyen and Venkatesh proposed the use of HHMM to recognize complex human movements [238,239]. Jalal et al. [240] extracted a multifused model for statistical, wavelet and binary features to maximize the occurrence of optimal feature values. The MEMM is used for further processing to obtain the empirical expectation value and the maximum entropy, which can measure the signal variance to achieve extraordinary accuracy results.

The design of activity/gesture recognition algorithms usually requires information about the position and orientation of the IMU on the body, and the signal processing is usually designed to be used in conjunction with the known sensor orientation and position. However, the sensor can be worn or held in different ways, so the performance of the signal processing algorithm may not be as expected. Wu et al. [241] proposed an orientation-independent activity/gesture recognition method by exploring a new feature set that works no matter how the sensor is oriented. This method uses dynamic time warping (DTW) for segmentation. Compared with traditional classification algorithms (such as support vector machines with large feature sets, Bayesian networks or hidden Markov models), it requires simple fixed-point operators and less computing resources. In addition, this model is suitable for all low-power wearable sensors, and the method can be implemented when the direction of the IMU changes. Experimental results show that the average accuracy of this method for daily life recognition and gesture recognition is 98.2% and 95.6%, respectively.

4.3. Deep learning algorithms

Deep learning is an end-to-end learning method. It starts directly from the original input data and automatically completes the process of feature extraction and model learning through a layered network. Deep learning not only avoids the tedious design of features by hand and makes the features of automatic learning have good effect, but also can use less parameters to fit more complex functions and express more complex models. It has strong advantages in feature expression, function modeling and model generalization. Fig. 7 shows the difference and relationship between deep learning and machine learning in the process of human activity recognition.

4.3.1. Deep belief network and restricted boltzmann machine

Deep belief network is a probabilistic graphical model that can effectively learn complex dependencies between variables. It contains many layers of hidden variables, so it can effectively learn the internal feature representation of the data and can also be used as a nonlinear dimensionality reduction method. Deep Belief Network solves the optimization problem of deep neural networks by using layer-by-layer training. The layer-by-layer training gives a good initial weight to the whole network, so that the network can reach the optimal solution with fine tuning. And the most important role in the layer-by-layer training is played by the “restricted Boltzmann machine”.

Yu et al. [149] applied a deep belief network to the field of Chinese sign language recognition based on wearable sensors, surface electromyography, accelerometer and gyroscope signals, and explored three different sensor fusion strategies including data-level fusion, feature-level fusion and decision-level fusion. The results show that feature-level fusion can achieve the best recognition accuracy among the three fusion strategies. Fang et al. [242] established a DBN with 4 hidden layers. The algorithm of restricted Boltzmann machine is used to pre-train layer by layer to solve the problem of human activity recognition. Compared with hidden Markov model and naive Bayes model, it is found that the deep belief network performs better. Hassan et al. [156] used DBN with two hidden layers and 100 inputs for activity recognition, and compared it with support vector machine and artificial neural network. The results show that the recognition performance of DBN is better than that of SVM and ANN. Oukrich et al. [243] applied three machine learning algorithms to represent and recognize human activities, and compared deep belief network with traditional recognition methods such as support vector machine and back propagation algorithm (BPA). The main reason is the suitability of DBN for human activity recognition and its strong ability to learn and interpret complex sensor events in a smart home environment.

4.3.2. Deep auto-encoder

Deep Auto-Encoder (DAE) is to learn an effective encoding (or representation) of a set of data in an unsupervised way. DAE is divided into two parts: encoder and decoder. The encoder unit converts the input data into hidden features, and the decoder part reconstructs the hidden features into approximate representations to reduce the possibility of error rate [244]. DAE methods have been developed for feature representation in smartphones and health monitoring applications [245]. Munoz-Organero et al. proposed a new generative model for training a group of auto-encoders to learn features that can detect human motion [246]. Zhou et al. [163] used a deep learning algorithm based on stacked auto-encoders for human motion recognition, which proved to be more accurate than traditional neural networks with higher motion recognition accuracy.

In recent years, people have proposed a variety of feature representation methods for automatic encoders. Stacked auto-encoders [247], deep denoising auto-encoders (DDAE) [248], and sparse auto-encoders are widely used in human activity detection based on feature fusion [244,249]. Sparse Auto-Encoder solves the problem of high-dimensional feature vector representation by introducing the sparse term loss function, learning sparse and over-complete data representation, and making it linearly separable. The advantage of the sparse autoencoder is that it has high interpretability and implicit feature selection at the same time.

In addition, we use DAE to get an effective data representation. Besides the minimum reconstruction error or sparsity, an effective data representation can also require other properties, such as robustness to partial data corruption. DDAE increases the robustness of coding by introducing noise. It uses partial destruction of the input samples to reconstruct the original input data trained in unsupervised layer by layer initialization to capture the robust data invariants to changes [250].

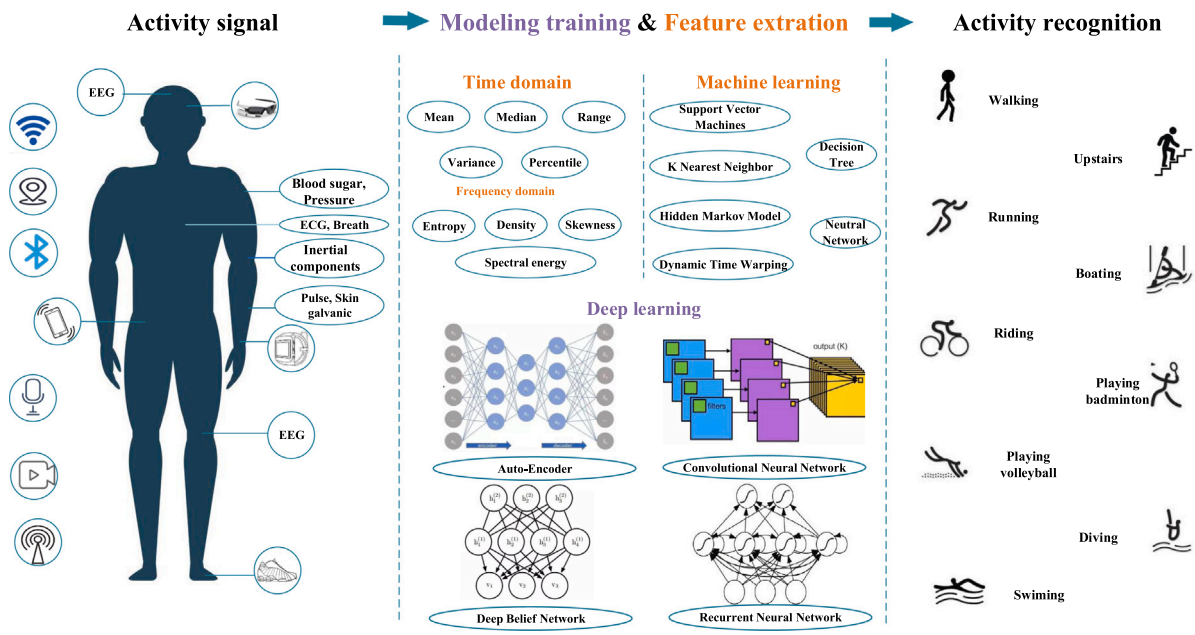

Fig. 7. The difference and connection between machine learning and deep learning for human activity recognition.

4.3.3. Convolutional neural network

Convolutional neural networks (CNNs) [26] have achieved good results in speech recognition [251], image classification [252], and video activity recognition [253]. When applied to human activity recognition related to time series classification, CNNs are superior to other traditional ML methods due to their local dependence and scale invariance [25].

In [153], a novel deep learning architecture based on a convolutional neural network is proposed for automated drowsiness detection using a single-channel EEG signal. Moniri et al. [254] used shallow models and deep convolutional neural networks (CNN) to learn and predict 5 common surface electromyography (sEMG) features simultaneously in real time. The results show that CNN can predict 25 s in advance, and the average absolute percentage error of all features is 6.88%, and the standard deviation of the absolute percentage error is 3.72%. In addition, CNN is at least 30% better than the best shallow models in terms of accuracy and precision. Jang et al. [185] used deep neural network and convolution neural network algorithm to process sensor data, so as to classify bad postures. Molchanov et al. [255] proposed a new multi-sensor system for accurate dynamic human gesture recognition, and used convolution depth neural network to fuse data from multiple sensors to classify human gestures.

In [256], the force plate and body-worn IMUs were used to capture movements, and random forest and convolutional neural network-based regression models were used to estimate participant-specific fatigue levels. Zhang et al. [106] proposed a new partial directed coherence graph convolution neural network (pdc-gcnn) model for EEG fatigue detection, which fused the features of time–frequency and brain topology.

Yang et al. [257] invented a new type of soft wear-resistant shoes for sports intention learning. This smart shoe consists of a soft sole with a pneumatic sensor and two 3-D motion sensors to obtain information about the ground reaction force and the foot sports information. Firstly, the multi-sensor data is constructed as “sensor image”, which can represent the space–time structure between datasets. Then, convolutional automatic encoder (CAE) is used to fuse multi-sensor datasets and extract features from 2-D images. Finally, LSTM can be used to understand the time evolution of human gait in the long run and to model the highly nonlinear input–output relationship in the short run. This method can fully understand the deep relationship between multiple sources of information, and learn people’s movement intentions at an unspecified walking speed.

4.3.4. Recurrent neural network

As a learning method and estimator, recurrent neural network (RNN) is suitable for processing time series data, such as audio [258] and text [259]. There are connections between the hidden layer units of RNN, which allow information to be transferred from one neuron to the next neuron, so that RNN can extract the time relationship. However, when the input sequence is too long, RNN is prone to gradient disappearance, which makes it lose the ability to connect long-distance information. In order to solve this problem, people have made a lot of improvements to the recurrent neural network, and the most effective way is to introduce gating mechanism, such as long short memory network (LSTM) and gated recurrent unit neural network (GRU).

(1) LSTM:

a. LSTM alone

Long Short-term Memory (LSTM) [260] is an neural network structure that aims to store and access information better than standard RNNs. LSTM has recently achieved the most advanced results in various sequence processing tasks, including gait evaluation [187,257], log anomaly detection [261], image recognition [262], action recognition [263,264], and stress detection [265]. Tran et al. [155] proposed a new multi-model LSTM network for learning the gait temporal features. Ullah et al. [157] proposed a stacked LSTM network for training with accelerometer and gyroscope data. These researchers found that using stack LSTM network to repeatedly extract temporal features can improve the recognition efficiency. Murad et al. [266] proposed a one-way, two-way and cascaded architecture of deep recurrent neural network based on LSTM, and proved that it is superior to traditional machine learning algorithms. On this basis, Inoue et al. [267] proposed a high-throughput human activity recognition method using DRNN for acceleration raw data. Guan et al. [268] combined multiple LSTM learners into an integrated classifier for the first time, which greatly improved the robustness of the human activity recognition system.

b. CNN+LSTM

At present, the method of combining convolutional neural network and long–short-term memory neural network for feature extraction has been widely used. Kavi [269] discussed the integration of LSTM architecture and CNN in multi-view applications. Hssayeni et al. [270] used a dual-channel, LSTM for hand-crafted features, 1D Convolutional Neural Network (CNN)-LSTM for raw signals, and 2D CNN-LSTM for time–frequency data. Ordonez [8] constructed a deep framework for

human activity recognition based on CNN and LSTM. The network includes an input layer, 4 one-dimensional convolutional layers, two consecutive LSTM layers, and a Softmax layer. Tora et al. [127] designed a deep convolutional neural network based on sensor-based activity classification to study the application of deep learning in beach volleyball movement recognition, and compared the performance of this new method with five common classification algorithms. It was found that the use of deep convolutional neural networks can achieve a classification accuracy of 83.2%, which is 16.0% higher than other classification algorithms. Mekruksavanch et al. [160] proposed a hybrid deep learning network composed of CNN layer and LSTM layer. The network has the ability of automatic learning space characteristics and time representation, and is suitable for human activity identification of sensor data of smart phones. Mutegeki et al. [159] combined the CNN network with the LSTM network on the basis of previous studies. It not only uses the robustness of CNN network feature extraction, but also uses the LSTM model to classify the time series to obtain better recognition performance.

(2) GRU:

The gated recurrent unit (GRU) network is a simpler recurrent neural network than the LSTM network. Chung et al. [271] comparatively studied the role of GRU and LSTM in human activity recognition, and found that the performance of the two is equivalent. In [272], a Deep Neural Network based model, which uses CNN, and GRU is proposed as an end-to-end model performing automatic feature extraction and classification of the activities as well. Yang et al. [273] proposed a region-convolutional neural network (RCNN) to identify the gait types of the wearer. For each gait type, the inertial data from the wearer's thighs and feet are collected at the same frequency, and the corresponding gated recurrent unit-support vector regression deep hybrid neural network model is constructed for virtual IMU simulation. Kim et al. [188] proposed a new arm gesture recognition method based on gyroscope and accelerometer sensors using deep convolution and recurrent neural networks. This method uses four deep convolution layers to automate feature learning in raw sensor data. The features of the convolution layers are used as input of the GRU which is based on the state-of-the-art RNN structure to capture long-term dependency and model sequential data.

4.4. Transfer learning

A prerequisite assumption for standard machine learning and deep learning is that the distribution of training and test data is the same. If this assumption is not satisfied, the model learned on the training set will perform worse on the test set. However, recording enough data for each new user is non-viable due to the limitations and challenges of working with human users. Transfer learning techniques aim to transfer the knowledge which has been learned from the source domain (subject) to the target domain in order to decrease the models' performance loss in the target domain [274].

The transfer learning technique is applied to deep neural network-based model to reduce the training effort [270,275–278]. Soleimani et al. [274] presented a novel method of adversarial knowledge transfer named SA-GAN stands for Subject Adaptor GAN, which utilizes the Generative Adversarial Network framework to perform cross-subject transfer learning in the domain of wearable sensor-based human activity recognition. Shickel et al. [279] overcomed the issue of small sample size in the intelligent intensive care unit cohorts by employing transfer learning techniques. Fu et al. [280] designed a compact wireless wearable sensor node, which combines an air pressure sensor and inertial measurement unit to provide multi-modal information for human activity recognition model training. To solve personalized recognition of user activities, they proposed a new transfer learning algorithm, which is a joint probability domain adaptive method with improved pseudo-labels (IPL-JPDA). Banos et al. [281] presented an alternate approach based on transfer learning to opportunistically train new unseen or target sensor systems from existing or source sensor systems. The method uses system identification techniques to learn mapping functions that automatically convert signals from the source sensor domain to the target sensor domain and vice versa.

To tackle this cross-domain HAR problem, Zhao et al. [282] proposed a domain adaptation (DA) method called local domain adaptation (LDA), whose core is to align cluster-to-cluster distributions between the source domain and the target domain. Specifically, firstly, the activity class is divided into several high-level abstract clusters. Secondly, the original data of each cluster in the two domains are mapped to the same low dimensional subspace to analyze the data distribution in the cluster. Finally, the class labels of the target domain in the low dimensional subspace are predicted. Ding et al. [283] found that Maximum Mean Discrepancy (MMD) is most suitable for human activity recognition by comparing several common algorithms. In addition, by studying the distribution characteristics of sensor data extracted by CNN, an improved method combining MMD with center loss is proposed.

Transfer learning leverages the labeled samples from the source domain to annotate the target domain which has few or none labels. Choosing the right source domain helps the algorithm perform well and prevents the negative transfer. Wang et al. [284] proposed an effective Unsupervised Source Selection algorithm for Activity Recognition (USSAR). USSAR is able to select the most similar K source domains from a list of available domains. In addition, they proposed an effective Transfer Neural Network to perform knowledge transfer for Activity Recognition (TNNAR). TNNAR could capture both the time and spatial relationship between activities while transferring knowledge. Khan et al. [285] proposed a transductive transfer learning model for CNN. Assuming that as long as the target activity set remains unchanged, the relative distribution of weights in different CNN layers remains unchanged. Silva et al. [286] proposed a method based on pre-trained encoders working as feature extractors.

5. Recent challenges and future research directions

At present, the theoretical research and practical application of wearable devices and information fusion have led to many more challenging research directions, as shown in Fig. 8 and Table 4. This paper selects the following three areas of interest to make a detailed discussion.

5.1. Recent challenges

(1) Hyper-parameters Optimization

In neural networks, there are many hyper-parameters in addition to the learnable parameters. These hyper-parameters also have a significant impact on the network performance. Different machine learning tasks often require different hyper-parameters. There are three common types of hyper-parameters as follows.

- Network structure, including the connection relationship between neurons, the number of layers, the number of neurons in each layer, the type of activation function, etc.
- Optimization parameters, including optimization methods [287–289], learning rate, number of samples in small batches and so on.
- Regularization factor.

There are two main difficulties in hyper-parameters optimization:

- Hyper-parameters optimization is a combinatorial optimization problem, which cannot be optimized by gradient descent methods like general parameters, and there is no one general and effective optimization method.

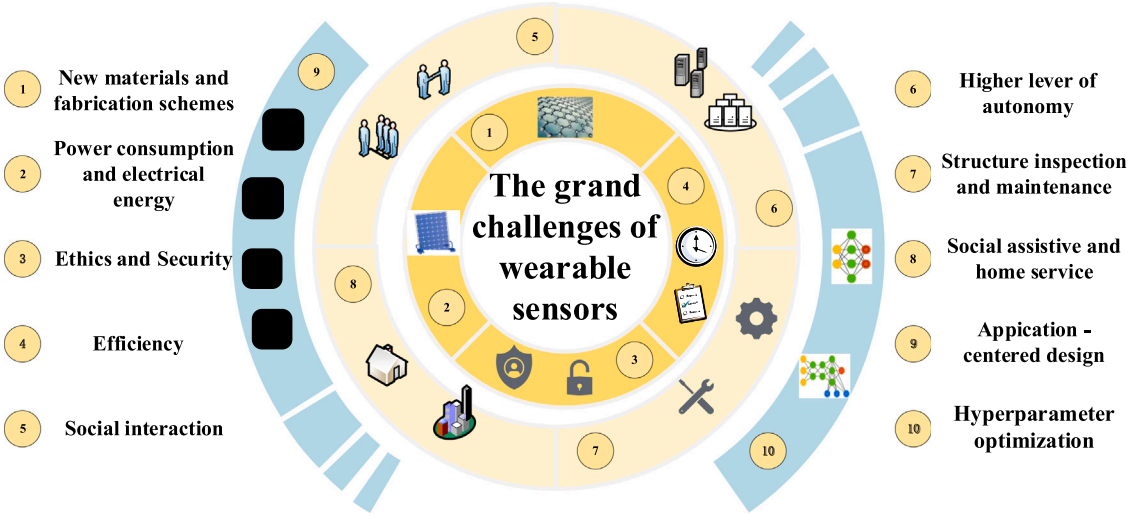

Fig. 8. Opportunities and challenges of the wearable sensor and information fusion.

Table 4
Recent challenges of human activity recognition.

Label	Challenges	Descriptations
1	Imbalanced data	Some sensor data of specific activities are challenging to obtain, such as those related to falls of elderly people. In addition, raw data recorded from unconstrained conditions is naturally class-imbalanced.
2	Distribution Discrepancy	The discrepancy between users which stems from different motion patterns when activities are performed by different people; data distributions of activities are changing over time, and new activities may also emerge; the discrepancy in sensors
3	Complex activities	Mixes complex and simple activities and tries to create a unified model; consider composite activities separately
4	Concurrent Activity	A person may carry out more than one activity at the same time
5	Multi-occupant Activity	Parallel activity where occupants perform activities individually and collaborative activity where multiple occupants collaborate together to perform the same activity
6	Computation Cost	The approach of improving model performance by introducing additional layers and nodes adds computational complexity
7	Privacy	Expose sensitive user information
8	Interpretability of data	The inner mechanisms of deep learning networks still remain unrevealed
9	Overfitting	Due to the large number of parameters involved and complex correlation, the DL methods are likely to lead to over-fitting of data in the training stage
10	Hyperparameter optimizations	Convolutional layers, filters, filter size, neurons, epochs, sliding window are certain parameters that every model needs to select the best value for better performances
11	Annotation Scarcity	It is more challenging to assign labels to a large amount of data

- The time cost of evaluating a set of hyper-parameters configurations is very high, which makes it difficult to apply some optimization methods in hyper-parameters optimization.

Take LSTM as an example. The LSTM unit contains a memory block and gate to let information pass through the LSTM connection. These doors have several in and out connections, and each door has its own parameters to be trained. In addition to these connections, there are other hyper-parameters, such as the number of hidden neurons and batch size. Therefore, the use of LSTM to achieve good or even the most advanced performance requires the selection and optimization of hyper-parameters, and there is no universal optimal configuration for all problem domains. Nakisa et al. [290] used differential evolution (DE) algorithm to automatically optimize LSTM hyper-parameters for the first time to identify and classify emotions. The results show that although DE spends the most processing time in all iterations, its performance is higher than all other algorithms. To enable fair comparison among the different DL architectures, Bayesian optimization was used for hyper-parameters tuning [291]. However, there are relatively few research results on hyper-parametric optimization.

(2) Privacy and security

The development of wearable medical devices with sensors has expanded the security issues of wearable networks through the enhancement of the Internet of Things. Major challenges can be summarized as follows.

- Due to the private and sensitive nature of personal data and the wireless nature of communications, the security of communications between these devices is critical to protect transmitted data from eavesdropping, tampering, or injection of malicious commands [292].
- Existing robust security solutions are not suitable for wearable devices. First, traditional security solutions proposed for computers are too computationally demanding for miniaturized wearable IoT devices [293]. Second, the recently emerged biometric encryption based on physiological biometrics has some shortcomings in terms of data collection, resistance to interference and resistance to attacks.

- As the cost and size of acceleration sensors decrease, biometric methods based on behavioral biometrics are gaining attention for their non-invasive and unique characteristics [30].

Hence, subsequent research is expected to apply a multi-biometric fusion-based security approach to further improve the security of wearable devices.

(3) Power consumption and electrical energy

To support multi-sensor devices for multi-parameter tracking and continuous operation, a robust energy supply unit that is compact, mobile, with high energy density and self-charging capability is required. Currently, there are only a few successful examples of self-powered (or wireless-charged) sensing systems that can operate for short periods of time (e.g., tens of seconds) after a sufficiently long recharge with an energy harvesting device (e.g., an organic photo voltaic cell [294] or triboelectric nanogenerator [295]). Yet, achieving a long operation duration for such systems remains to be a grand challenge. There are mainly the following aspects:

- The data processing and transmission units of the system consume most of the emergy. Therefore, it is necessary to implement a sensor with high sensitivity and low noise to reduce the workload of the signal acquisition/preprocessing circuit and microcontroller unit (MCU) and save energy.
- Due to the rapid development of energy harvesting technologies, flexible energy harvesting units, combining flexible photovoltaics, thermoelectric generators, and triboelectric nanogenerators that harvest ambient light energy, humanbased thermal energy, and mechanical energy, can already generate sufficient energy to support very frequent measurements of vital signs [296]. Hence, it is useful to construct a self-powered system with a simple structure.
- To achieve the ultimate goal of a self-powered continuous vital signs monitoring system, new energy harvesting technologies need to be explored and the power conversion efficiency of existing technologies needs to be improved through material and device innovations.

In addition, even though we have achieved record performance energy devices, we still have much work to do to improve the scalability, stability and repeatability of our energy devices in order to bring wearable self-powered systems to commercialization and widespread adoption.

5.2. Future research directions

(1) Unsupervised learning

Unsupervised learning to features from a given dataset remains an important research direction in current artificial intelligence. Existing unsupervised learning methods are mainly based on two mainstream generative models: deep belief networks (DBN) and autoencoders. They consist of multiple layers of hidden units that can be used to extract features and find patterns in massive amounts of data. In addition, for the convenience of readers, we have put the the abbreviations of detailed descriptions of methods and sensors in a separate table, as shown in Table 5.

Barbosa et al. [297] investigated the use of Transfer Learning, specifically Unsupervised Domain Adaptation, within human activity recognition systems. Islam et al. [298] proposed unsupervised transfer learning from source dataset to target dataset. They used Maximum Mean Discrepancy (MMD) based transfer learning model and compared with base CNN model. Xu et al. [299] proposed an automatic data collection and labeling system in order to solve the annotation problem. And by combining supervised and unsupervised machine learning methods, a novel coarse-to-fine activities of daily living (ADLs) recognition algorithm for elderly people is gived. Multiple sensors fusion strategy is employed to interpret and annotate the ADLs, the recognition accuracy is larger than 91%. Abedin et al. [300] noted that collecting and annotating sensory activity datasets is costly, very time consuming, and does not scale to large amounts of data. Therefore an unsupervised end-to-end learning strategy was developed for the fundamental problem of human activity recognition for wearable devices.

Ma et al. [301] proposed an end-to-end multi-task deep clustering framework. Using unlabeled multidimensional sensory signals as input, a convolutional neural network-bidirectional Long Short-Term Memory (CNN-BiLSTM) self-encoder is used to form a compressed latent feature representation. Then a feature extraction-based K-means clustering algorithm is used to divide the dataset into different groups and generate pseudo-labels for the instances. Further, a deep neural network is trained with the latent features and pseudo-labels for activity recognition. The tasks of feature representation, clustering and classification are integrated into a unified multi-task learning framework and jointly optimized for unsupervised activity classification. Experimental results based on three public datasets show that the method outperforms shallow unsupervised learning methods and is fine-tuned to approach state-of-the-art supervised learning methods with a small amount of labeled data. The paper significantly reduces the cost of human-based data labeling and closes the gap between unsupervised and supervised human activity recognition. To address the challenge that transfer learning requires access to the original dataset which may lead to privacy leakage, Hashemian et al. [302] proposed centralized and distributed semi-supervised and unsupervised transfer learning scenarios. And the recognition error rate is reduced by about 20.58% and 15.97%, respectively, while privacy is protected to a large extent.

(2) Data augmentation

Data augmentation techniques are a possible solution for those whose goal is to increase the amount of available data and avoid overfitting problems. Data augmentation consists of methods to improve the attributes and size of the training datasets, capable of solving data space problems for limited-data.

Steven et al. [303] identified the lack of labeled data as a barrier factor in understanding the performance of data-driven learning models. To cope with this problem, a feature space-based data augmentation integration is proposed. And the impact of feature representations and augmentations on activity recognition accuracy is explored through performance tests on deep long and short-term memory neural network structures. Shorten et al. [304] covered a large number of applications of enhancement methods based on generative adversarial networks, and discussed other characteristics of Data Augmentation such as test-time augmentation, resolution impact, final dataset size, and curriculum learning.

Kim et al. [305] used sensors to predict the magnitude of impact acceleration after loss of equilibrium. To improve the prediction performance, a data augmentation technique with increment of dataset is also applied. Elnaggar et al. [306] proposed a wearable device based human posture classification method which can classify 12 baseline sleeping postures. And a new data enhancement technique is proposed for the lack of inertial data to generate real synthetic datasets emulating real-world conditions. Jiang et al. [307] employed a new wearable sensor data enhancement method in order to develop algorithms that can accurately estimate motion fatigue. A large amount of simulated wearable human motion data is generated on the basis of a small corpus of human motion data measured by optical sensors.

(3) Attention mechanisms

The attention mechanism is an emerging technique that dynamically adjusts the focus of a neural network by scaling features using corresponding weights computed by independent attention modules. Given an appropriate attention design, the network can automatically scale the impact of informative features and suppress irrelevant noise.

Chen et al. [308] proposed a multi-agent spatial–temporal attention model which helps intelligently select informative modalities and their active periods. Liu et al. [309] introduced a lightweight attention mechanism called the global attention module for multi-sensor information

Table 5
The following abbreviations are used in this manuscript.

Abbreviations	Terms	Descriptions
EMG	Electromyography	Muscle contraction generates a weak electric current, which can be measured by attaching electrodes to the skin at appropriate locations on the surface of the body. The curve of current intensity over time is called electromyogram
ACC	Accelerometer	Instrumentation for measuring the linear acceleration of a carrier body
GYR	Gyroscope	Angular velocity sensor for detecting angular velocity
MAG	Magnetometer	Measure the strength and direction of the geomagnetic field
PPG	Photoplethysmography	It can detect blood volume changes in microvessels and can be used for heart rhythm measurement, blood pressure measurement, etc
GSR	Galvanic skin response	Instrument for recording changes in electrical skin response
DT	Decision tree	A tree structure where each internal node represents a test on an attribute, each branch represents a test output, and each leaf node represents a category
KF	Kalman filter	Also known as linear Gaussian Markov model, its state space as well as observation space are continuous
EKF	Extended Kalman filter	The transformation between states and the transformation of states to observations is nonlinear
NB	Naive Bayes	One of the classical machine learning algorithms and one of the few classification algorithms based on probability theory
HMM	Hidden Markov models	Dealing with sequence labeling problems or tagging problems
MAP	Maximum a Posteriori	The biggest difference with maximum likelihood estimation is that MAP incorporates the probability distribution of the model parameters themselves
RNN	Recurrent neural networks	Processes sequence data, and the current output of a sequence is related to the previous output
DBN	Deep belief network	A generative model that allows the entire neural network to generate training data with maximum probability by training the weights among its neurons
ANN	Artificial neural networks	A nonlinear, adaptive information processing system consisting of a large number of interconnected processing units
PCA	Principle component analysis	Reduce the dimensionality of the dataset while maintaining the features that contribute the most to each other's differences in the dataset
CNN	Convolutional Neural Network	A class of feedforward neural networks that include convolutional computation and have a deep structure, which is one of the representative algorithms of deep learning
SVM	Support vector machine	A generalized linear classifier for binary classification of data by supervised learning
RF	Random forest	An integrated algorithm consisting of decision trees. New input samples enter and let each decision tree in the forest judge and classify them separately
KNN	k-nearest neighbors	A classification algorithm that discriminates a sample as the most similar to the K samples in the dataset
LSTM	Long Short-Term Memory Network	A special RNN that learns long-term dependencies
GRU	Gated recurrent unit	GRU is a simplified version of LSTM, a variant of LSTM, which uses hidden states for information transfer
SAE	Sparse auto-encoders	The most important feature of unsupervised neural networks is that the number of nodes in the input layer (excluding bias nodes) and the number of nodes in the output layer are the same, while the number of nodes in the hidden layer is less than the number of nodes in the input and output layers

fusion. It can use the information collected from the higher levels of the neural network to selectively amplify the impact of information features and suppress extraneous noise in the fusion layer. Ghazaleh et al. [310] used an attention-based recurrent neural network to capture the temporal features of hand motion to unveil the underlying pattern between the gesture and these sequences.

6. Conclusion

The advantage of wearable device is that it can be close enough to the user, so it can not only determine the user's behavior, but also determine the user's personalized behavior and cognitive data, which helps to identify the user's intention. This paper reviews the research status of wearable devices and information fusion.

In the aspect of wearing position, the position of the sensor has a direct impact on the motion measurement, and the best position of the sensor is determined by the user's acceptance and action classification accuracy.

Traditional machine learning algorithm has fast computing speed and good performance in simple action recognition, but it has the following limitations:

- Limited ability to express characteristics. Manual feature extraction is not only time-consuming and cost-effective, but also difficult to represent complex actions, so it is not suitable for complex scenes.
- The ability of modeling is limited, the model is relatively simple, and excessively depends on data features. The result of model classification depends largely on the quality of features, not the performance of the model itself.
- Limited generalization capability. Specifying the parameters of the model based on the task at hand is constrained in its ability to generalize to complex problems.

In general, deep learning provides flexibility, robustness and improved performance by harnessing the power of multilayer neural

networks. Compare with traditional manual feature extraction methods, deep learning can reduce feature design efforts by constructing end-to-end neural networks to automatically learn and acquire higher-level, more meaningful features, and deep neural network structures are better suited for online machine learning and unsupervised learning with better scalability. And the disadvantages are as follows.

- Since the target users have specific behavioral patterns, the recognition performance of the model will be significantly degraded when the behavioral data of the users differ significantly from the training dataset of the model.
- In practical application, there are a large number of target users. It is unrealistic to collect a large amount of tag data for each user, especially for the elderly, children and disabled people, and retrain a personalized model.

Some scholars have compensated for the above deficiencies by combining deep learning with transfer learning. Transfer learning techniques can improve the learning of a given task by transferring knowledge from related tasks, allowing training and testing data with different distributions, tasks and representations.

Finally, we point out unresolved issues and directions in research progress and suggest open research challenges that need to be brought to the attention of researchers.

CRediT authorship contribution statement

Sen Qiu: Conceptualization, Writing – original draft, writing – review & editing. **Hongkai Zhao:** Conceptualization, Writing – original draft, writing. **Nan Jiang:** Methodology, Review & editing. **Zhelong Wang:** Methodology, Review & editing. **Long Liu:** Methodology. **Yi An:** Conceptualization. **Hongyu Zhao:** Conceptualization. **Xin Miao:** Investigation. **Ruichen Liu:** Investigation. **Giancarlo Fortino:** Conceptualization, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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